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I. INTRODUCTION

With the rapid advancement of the Internet of Things (IoT) and 5G/6G communication technologies, the volume of data generated at the network edge has experienced explosive growth. To meet the demands of time-sensitive applications, such as autonomous driving and smart manufacturing, Edge AI [1], [2] has emerged. It focuses on deploying deep learning models as lightweight microservices on edge nodes in close proximity to data sources. However, practical edge computing environments face significant dynamic challenges. On the one hand, the physical resources of heterogeneous edge nodes (e.g., computational capacity and memory) are strictly constrained. On the other hand, a massive number of concurrent AI requests with stringent Service Level Agreements (SLAs) frequently emerge within the network. Such highly dynamic and time-varying traffic fluctuations necessitate that edge networks possess the capabilities for on-demand model orchestration and dynamic microservice scheduling [3], [4].

To provide high-performance AI services adapted to the physical constraints of edge nodes, existing cloud-edge orchestration schemes typically focus on jointly optimizing microservice deployment and request routing. For instance, some works emphasize energy-delay trade-offs in resource-constrained edge environments [5], [6], while others explore service mesh orchestration for large-scale and symmetric cloud systems [7], [8]. Furthermore, collaborative routing strategies have been proposed to handle dynamic call graphs and fluctuating traffic [9]. However, these approaches often assume static AI models and frequently treat AI services simply as fixed microservices [10], neglecting their architectural adaptability under varying resources. While Neural Architecture Search (NAS) [11] and supernet paradigms [12], [13] offer massive candidate subnets to overcome this, they suffer from severe response lag in online environments due to computationally expensive gradient-based evaluations. To circumvent this overhead, current frameworks often degrade to static matching based on offline finite model pools. However, these coarse-grained pools struggle to handle time-varying bursty traffic, failing to balance queueing delay and inference accuracy. Although Zero-Cost Proxies (ZCP) [14], [15] enable millisecond-level accuracy quantification, they remain confined to the pure architecture search domain. Crucially, they lack effective coupling with system-level dynamic microservice reloading, hard physical constraints, and queueing dynamics.

To address the conflict between the prohibitively high search overhead and the requirement for dynamic edge deployment, this paper proposes a cloud-edge collaborative deployment framework based on ZCP and dynamic scheduling. Specifically, this framework pre-trains a SuperNet offline in the cloud, while the edge triggers on-demand sampling and rapid evaluation mechanisms upon encountering traffic fluctuations. Leveraging ZCP, the system quantifies the accuracy potential and physical overhead of candidate subnets at the millisecond level. Building upon this, the edge scheduler executes hard-constraint filtering based on the transient available memory and computational upper bounds, and utilizes a dynamic heuristic algorithm to sense traffic variations, thereby adaptively adjusting the weights of soft constraints. Ultimately, the edge selects the optimal subnet and fetches its corresponding weights from the cloud on demand to complete rapid hot-reload deployment.

The main contributions of this paper are summarized as follows:

- We enhance an ultra-fast ZCP to eliminate time-consuming backpropagation, effectively decoupling architecture evaluation from the scheduling process. This enables millisecond-level performance estimations on edge nodes, saving critical time for model transmission and reloading.
- We propose a hard-constraint-driven dynamic heuristic scheduling algorithm. It filters subnets via real-time memory and compute limits, and adaptively tunes soft constraints based on concurrent traffic. This ensures proper degradation and sustains overall throughput during traffic bursts.
- We construct a cloud-edge decoupled lightweight

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deployment process. Edge nodes run only a minimal ZCP and parser to fetch subnets on demand. This eliminates local supernet storage, significantly reducing resource burdens while maintaining high service availability.

II. SYSTEM MODEL AND PROBLEM FORMULATION

This section proposes a cloud-edge collaborative on-demand microservice orchestration framework for delay-sensitive applications. This framework offloads the computation-intensive SuperNet pre-training to the cloud, while delegating the lightweight on-demand sampling, evaluation, and dynamic scheduling to the edge.

A. System Model

Suppose the system consists of a cloud center and a set of heterogeneous edge nodes denoted by $\mathcal{N} = \{n_1, n_2, \ldots, n_N\}$. The cloud offline pre-trains a supernet $\mathcal{W}$ containing a massive number of candidate subnets for the edge to fetch on demand. At any scheduling slot t, the transient physical available resource constraints of edge node $n_i \in \mathcal{N}$ are defined by a tuple: $Node_i(t) = \langle M_i^{max}(t), C_i^{max} \rangle$, which represent the currently available memory and the peak computational capacity (measured in FLOPS), respectively. Specifically, let M_{total} denote the total memory capacity of node n_i, and $M_{used}(t)$ denote the memory currently occupied by background processes and other multi-tenant services. The transient available memory is thus strictly bounded by $M_i^{max}(t) = M_{total} - M_{used}(t)$. When the system senses traffic fluctuations and triggers scheduling, the edge performs heuristic sampling on the supernet $\mathcal{W}$ and rapidly abstracts the candidate subnet a_k into a triple feature representation via the ZCP and parser:

$$a_k = \langle S_k^{proxy}, P_k^{model}, F_k^{flops} \rangle \tag{1}$$

where S_k^{proxy} is the theoretical accuracy potential score evaluated by ZCP; P_k^{model} and F_k^{flops} denote the number of parameters (memory requirement) and the number of floating-point operations (computational demand) of the subnet, respectively. To guarantee the time sensitivity of microservices under highly dynamic time-varying traffic $\lambda_i(t)$, the system sets an end-to-end delay SLA (denoted as T_{SLA}). For subnet a_k, its service rate on node n_i is approximated as $\mu_{i,k} = \frac{C_i^{max}}{F_k^{flops}}$. While neural network inference time on isolated hardware is typically deterministic, practical edge nodes often exhibit service time variability due to OS-level context switching, resource contention, and concurrent multi-tenant workloads. To capture this stochasticity while maintaining analytical tractability, we adopt the M/M/1 queueing model, which provides a conservative approximation for the system's average response time compared to deterministic execution. Thus, the average response time at slot t must satisfy:

$$T_{req} = \frac{1}{\mu_{i,k} - \lambda_i(t)} = \frac{1}{\frac{C_i^{max}}{F_k^{flops}} - \lambda_i(t)} \leq T_{SLA} \tag{2}$$

Consequently, the upper bound of the maximum tolerable computation $F_i^{max}(t)$ that the node can allocate to the target subnet is dynamically compressed by the traffic, which is derived as:

$$F_i^{max}(t) = \frac{C_i^{max}}{\lambda_i(t) + \frac{1}{T_{SLA}}} \tag{3}$$

Equation (3) establishes the computational limit from a physical boundary perspective: when the concurrent traffic $\lambda_i(t)$ surges, the upper bound of available computation shrinks sharply, and the system is forced to filter out heavy networks and select lightweight subnets with smaller F_k^{flops}, thereby mitigating the exacerbation of queueing congestion.

B. Problem Formulation

Let $x_{i,k}(t) \in \{0, 1\}$ be a boolean decision variable indicating whether subnet a_k is selected and fetched for node n_i at slot t (the total size of the candidate pool is denoted as K). The global scheduling problem is modeled as a constrained online joint optimization problem:

$$\max_{\mathbf{X}} \quad \sum_{t=1}^{T} \sum_{i=1}^{N} \sum_{k=1}^{K} x_{i,k}(t) \cdot \widehat{S}_k^{proxy} \tag{4}$$

$$\text{s.t.} \quad \sum_{k=1}^{K} x_{i,k}(t) = 1, \quad \forall i \in \mathcal{N}, \forall t \tag{5}$$

$$\sum_{k=1}^{K} x_{i,k}(t) \cdot P_k^{model} \leq M_i^{max}(t), \quad \forall i \in \mathcal{N}, \forall t \tag{6}$$

$$\sum_{k=1}^{K} x_{i,k}(t) \cdot F_k^{flops} \leq F_i^{max}(t), \quad \forall i \in \mathcal{N}, \forall t \tag{7}$$

Here, the objective function (4) aims to maximize the accuracy potential of the selected models. Constraint (5) guarantees that only one optimal architecture is activated per node at a time. Constraints (6) and (7) constitute the hard-constraint filtering pool, strictly ensuring that the overhead of the selected subnet does not exceed the node's transient available memory and the queueing computational upper bound defined by Equation (3).

III. ONLINE ZERO-COST ARCHITECTURE EVALUATION

To enable the cloud-edge collaborative pipeline of "edge search, cloud delivery, and edge deployment" to meet strict real-time requirements, the system must perform ultra-fast multi-dimensional attribute evaluation of candidate architectures at the edge, thereby generating the attribute tuple $a_k = \langle S_k^{proxy}, P_k^{model}, F_k^{flops} \rangle$ that encapsulates accuracy potential and resource overhead. Since the

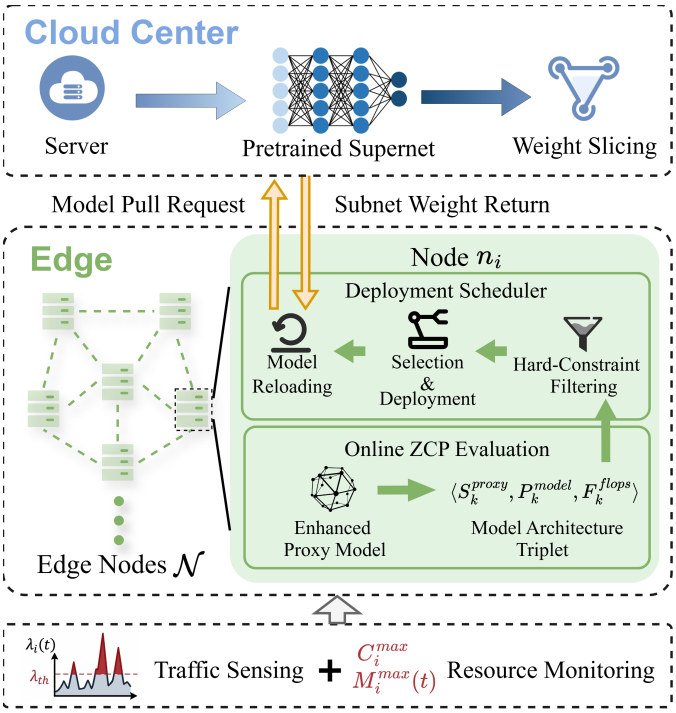

Fig. 1. System architecture of the proposed cloud-edge collaborative deployment framework based on ZCP and dynamic scheduling. The cloud center pre-trains the supernet offline, while the edge node relies on traffic sensing and resource monitoring to perform online ZCP evaluation and heuristic scheduling, ultimately reloading the optimal subnet on demand.

computation volume (F_k^{flops}) and parameter size (P_k^{model}) of the candidate subnet a_k are constant values once its operator-level topology is determined, the edge scheduler only needs to conduct a lightweight static topology traversal with $\mathcal{O}(L)$ complexity to deterministically obtain these two physical overhead features. Therefore, this section primarily focuses on how to circumvent expensive online gradient training and achieve accurate model accuracy potential evaluation (S_k^{proxy}) tailored to heterogeneous edge scenarios under ultra-low latency.

Traditional zero-cost proxies often adopt static, unified metrics, making it difficult to adapt to the complex and variable hardware constraints and traffic fluctuations in edge computing environments. To this end, this framework introduces the GSTC proxy model [15] as the zero-cost proxy and proposes a dynamic adaptation mechanism tailored for heterogeneous edge states.

After the edge scheduler generates the Directed Acyclic Graph (DAG) for the candidate subnet a_k, the evaluator utilizes the locally cached operator information retention rate (η) to efficiently extract four fundamental physical quantities characterizing the information flow of the network topology: the total architectural path information G, the path information difference S, the total information received by the output node T, and the information difference received by the output node C. To ensure that the proxy evaluation naturally aligns with the transient resource constraints and service demands of the current edge node, the system dynamically allocates weights $(\alpha, \beta, \gamma, \delta)$ to each component, or even performs dimensional pruning, to calculate a customized proxy score:

$$S_k^{proxy} = \alpha \cdot G + \beta \cdot S + \gamma \cdot \widehat{T} + \delta \cdot \widehat{C} \tag{8}$$

where $\widehat{T}$ and $\widehat{C}$ are min-max normalized terms based on the upper and lower bounds of the global search space. Thanks to the high scalability of this evaluation mechanism, the scheduler can dynamically map to the most suitable evaluation scheme based on the node's real-time multi-dimensional states. The following lists three of the most representative typical scenarios:

Low-Latency & Resource-Constrained Scheme: When edge nodes face extreme physical resource constraints or are experiencing peak concurrent traffic, complex network topologies, despite enhancing information difference, can lead to severe memory fragmentation and unpredictable memory access latency. In such cases, the system triggers a dimensionality reduction and pruning strategy, setting the weights of the structural difference terms to zero (i.e., $\beta = \delta = 0$) and potentially introducing a negative decay penalty to the total information G. This scheme eliminates the incentive mechanism for network complexity, guiding the scheduler from the evaluation source to favor candidate architectures with shallower layers and simpler topologies, thereby maximizing the system throughput and overload resistance.

High-Accuracy & Relaxed-Latency Scheme: For nodes with relatively abundant computational resources and stringent requirements for inference accuracy, the primary optimization objective shifts to enhancing the representation capability of the model. The physical quantities G and $\widehat{T}$ are highly positively correlated with the network depth and feature extraction capacity. Therefore, the system assigns exceptionally high weight factors to the total information and the output information (i.e., $\alpha, \gamma \gg 1$). This scheme can capture high-complexity subnets within the search space that exhibit lossless information flow and extremely deep hierarchical structures, ensuring the maximum utilization of the underlying hardware computational capacity within resource constraints to trade off ultimate accuracy.

Balanced Robust Scheme: When node resources and traffic are in a steady state, the system adopts the full set of evaluation components with balanced weights ($\alpha = \beta = \gamma = \delta = 1$). This scheme jointly considers network depth (characterized by G and T) and the diversity of residual structures (characterized by S and C), achieving an optimal comprehensive trade-off between theoretical accuracy potential and structural redundancy to provide a highly robust normalized candidate set.

Through the aforementioned dynamic adaptation mechanism, the system ingeniously pre-decouples the compu-

tationally heavy online joint optimization problem. Edge nodes can generate a multi-dimensional set of candidate architectures within milliseconds that not only comply with physical constraints but also possess topological characteristics naturally suited to transient traffic scenarios. This process secures a substantial time margin for the subsequent on-demand fetching of models from the cloud.

IV. CLOUD-EDGE COLLABORATIVE HEURISTIC DEPLOYMENT

Upon triggering a scheduling event, the edge scheduler first evaluates the transient resource state (e.g., $M_i^{max}(t)$) and concurrent traffic $\lambda_i(t)$ to dynamically configure the GSTC proxy weights $(\alpha, \beta, \gamma, \delta)$ according to the context-aware schemes outlined in Section III. By executing this lightweight static topology traversal, the system generates the customized accuracy potential score S_k^{proxy} for each candidate. Based on these extracted context-aware attributes, the edge scheduler then executes a dynamic deployment pipeline comprising three stages: hard-constraint filtering, heuristic weighted selection, and on-demand fetching. This two-tier dynamic mechanism copes with bursty traffic micro-states and realizes fine-grained adaptive proper degradation.

First, in the physical constraint filtering stage, the engine processes the candidate subnet set $\mathcal{A}$ in $\mathcal{O}(|\mathcal{A}|)$ time based on transient resource bounds. It eliminates subnets exceeding the memory limit ($P_k^{model} > M_i^{max}(t)$) to prevent Out-of-Memory (OOM) failures, and those violating the queueing computation limit ($F_k^{flops} > F_i^{max}(t)$) to strictly guarantee SLA delay boundaries under the transient arrival rate $\lambda_i(t)$. This safely prunes the global search space into a feasible real-time subset $\mathcal{A}_i^*(t)$.

Subsequently, the process enters the dynamic heuristic degradation stage. To optimally trade off accuracy potential, inference latency, and memory footprint within $\mathcal{A}_i^*(t)$, we design a utility function based on Global Boundary Normalization:

$$U(a_k) = w_1(t) \cdot \widehat{S}_k^{proxy} - w_2(t) \cdot \widehat{F}_k^{flops} - w_3(t) \cdot \widehat{P}_k^{model} \tag{9}$$

where $\widehat{(\cdot)}$ denotes min-max normalization over the entire cloud supernet to prevent local scale distortion, and $\widehat{S}_k^{proxy}$ is the theoretical accuracy score evaluated by our customized GSTC. To handle microscopic fluctuations, the penalty weights are dynamically driven by transient traffic $\lambda_i(t)$ and resource utilization:

$$\begin{aligned} w_2(t) &= c_1 \cdot \exp\left(\max\left(0, \frac{\lambda_i(t) - \lambda_{th}}{\lambda_{th}}\right)\right), \\ w_3(t) &= c_2 \cdot \frac{M_{used}(t)}{M_{total}} \end{aligned} \tag{10}$$

with the normalized accuracy weight $w_1(t) = \frac{1}{1+w_2(t)+w_3(t)}$, where λ_{th} is a pre-calibrated safe traffic threshold. This normalized weight distribution intrinsically forms a self-adaptive penalty mechanism. When traffic is stable ($\lambda_i(t) \leq \lambda_{th}$), $w_2(t)$ remains constant, prioritizing the GSTC-evaluated accuracy. Conversely, during traffic bursts ($\lambda_i(t) > \lambda_{th}$), $w_2(t)$ grows exponentially, severely penalizing computation and suppressing $w_1(t)$. This proactively triggers system-level degradation, trading accuracy for lightweight architectures to sustain high throughput and service availability.

Finally, in the on-demand fetching stage, the scheduler identifies the optimal subnet $a^* = \arg\max_{a_k \in \mathcal{A}_i^*(t)} U(a_k)$ and sends its Architecture Encoding to the cloud. The cloud server instantly slices the corresponding weights from the pre-trained supernet $\mathcal{W}$ and transmits them to the edge. By bypassing the massive gradient training overhead via the GSTC proxy, the entire "search-decision-fetch" duration is strictly compressed, enabling agile hot-reload microservice deployment within a tolerable time window.

Algorithm 1 Cloud-Edge Collaborative Dynamic Heuristic Scheduling

Require: Supernet candidate set $\mathcal{A}$, transient state $M_i^{max}(t), \lambda_i(t)$, SLA bound T_{SLA}, safe threshold λ_{th}
Ensure: Optimal subnet encoding a^* for current time slot t.

```
Determine GSTC weights (α, β, γ, δ) based on transient resource limits and traffic;
for each candidate a_k ∈ A do
    Calculate customized proxy score S_k^proxy via Eq. (8);
    Extract physical attributes to form tuple a_k = ⟨S_k^proxy, P_k^model, F_k^flops⟩;
end for
Calculate dynamic compute limit F_i^max(t) via Eq. (3);
Initialize feasible subset A_i^*(t) ← ∅;
for each candidate a_k ∈ A do
    if P_k^model ≤ M_i^max(t) AND F_k^flops ≤ F_i^max(t) then
        A_i^*(t) ← A_i^*(t) ∪ {a_k};
    end if
end for
if A_i^*(t) is empty then
    return System Failure / OOM
end if
Calculate penalty weights w_2(t), w_3(t), and normalized weight w_1(t) via Eq. (10);
for each candidate a_k ∈ A_i^*(t) do
    Normalize attributes to Ŝ_k^proxy, F̂_k^flops, P̂_k^model ∈ [0, 1];
    Calculate utility score U(a_k) via Eq. (9);
end for
Select optimal architecture: a* = arg max_{a_k} U(a_k);
Send a* encoding to the cloud server;
Cloud slices weights from W and edge hot-reloads a*;
return a*
```

V. EXPERIMENTAL ANALYSIS

A. Experimental Settings

Model library. We pre-train a SuperNet $\mathcal{W}$, forming a multi-version model library of 80 task types. Each subnet is represented by $\langle S_k^{proxy}, P_k^{model}, F_k^{flops} \rangle$ in Eq. (1), with inference accuracy normalized to QoS in $[0.1, 1.0]$.

Edge setup. The edge network contains $N = 3$ nodes by default, scaling to 5 and 7 in the scale experiment. Each node provides $C_i^{max} = 200$ GFLOPS and $M_{total} = 150$M parameters, with a fixed cross-node delay of $\tau_{comm} = 0.02$s.

Traffic. User requests follow pipelined call chains [?] with lengths from 3 to 10. The arrival rate ranges from 100 to 800 requests/s, and per-chain rates are drawn from a uniform Dirichlet distribution with $\alpha = 1$.

Baselines. Six baselines are compared: First-Fit Decreasing (FFD), Chain-aware Deployment Scoring (CDS), Random Local Search (RLS), Greedy (Greedy), Dynamic Resource Scaling (DRS), and LEGO [?]. All baselines deploy a fixed model per task, while the proposed algorithm dynamically selects models via hard/soft constraints. The proposed algorithm runs 40 generations with population 36, $\omega = 5.0$, and a QoS-delay tradeoff of 1.2.

B. End-to-End Latency

The hard constraint $F_i^{max}(t)$ in Eq. (3) compresses the allowable computation budget as traffic increases, forcing the scheduler toward lighter subnets. This keeps end-to-end latency far below all baselines (Fig. 2) across every perturbation axis.

When the arrival rate rises from 200 to 800 req/s (Fig. 2(a)), FFD grows from 0.37 s to 3.39 s and the other baselines fall in the range 0.35–0.43 s at $\lambda = 200$ and 2.8–3.4 s at $\lambda = 800$. Our method grows from 0.13 s to 1.14 s. At $\lambda = 400$ its average delay is 0.14 s, 93% below the baseline mean of 2.02 s.

Under chain-length perturbation (Fig. 2(c)), the gap widens further. At length 8, baselines reach 3.5–4.0 s due to cascading queueing across hops, whereas our method stays at 0.23 s. The exponential penalty $w_2(t)$ in Eq. (10) grows large enough to override $w_1(t)$; the scheduler therefore trades inference quality for queue stability. The task-type axis (Fig. 2(b)) shows a similar pattern: at $K = 10$ the proposed algorithm records 0.68 s versus 5.33 s for FFD, an 86% gap. Each node independently selects a subnet via Eq. (8) matched to its local resource state, rather than deploying a fixed model.

C. Resource Utilization

Fig. 3 shows the average memory utilization under the same three axes. The task-type perturbation (Fig. 3(b)) produces the widest gap: at $K = 5$ the proposed algorithm uses 97.3% of memory versus only 63.7% for baselines. Baselines spread a small number of fixed models across nodes and under-utilize memory, whereas the proposed algorithm selects larger subnets up to the hard memory constraint. At $\lambda = 200$ (Fig. 3(a)), it fills 94.0% of available memory, 9.6 percentage points above the baseline average of 84.4%. Since $w_2(t)$ stays small under light traffic, the utility function in Eq. (9) favors higher-accuracy (and thus larger) subnets within $M_i^{max}(t)$. As the rate climbs to 500 req/s, baselines catch up (96–98%) because their fixed models occupy most of the node capacity, while our method holds at 94.9%, leaving headroom for traffic bursts. Under chain-length variation (Fig. 3(c)), utilization stays above 95% for all algorithms at longer chains, as the increased hop count forces more deployments per node.

D. Scalability

Fig. 4 reports the fitness distribution over 100 independent trials at three network sizes. Because each node runs an independent ZCP evaluation and heuristic schedule, coordination overhead does not grow with N. The proposed algorithm achieves higher fitness at all three sizes: median 2.41 (Small, $N = 3$), 3.33 (Medium, $N = 5$), and 3.94 (Large, $N = 7$), versus the best baseline values of 1.74, 2.09, and 2.84 respectively. The relative advantage grows from 38% at Small to 59% at Medium, then drops to 39% at Large. This drop is expected: with more nodes, baselines also gain from shorter average routing paths, so the gap narrows.

Fitness increases with scale for all algorithms because more nodes spread the routing load and reduce queueing. The proposed algorithm gains more because each node evaluates and schedules independently. The cloud-edge split keeps per-node storage constant: a new node stores only the lightweight parser and proxy scorer locally, while the supernet remains in the cloud.

VI. CONCLUSION

ACKNOWLEDGMENT

The preferred spelling of the word “acknowledgment” in America is without an “e” after the “g”. Avoid the stilted expression “one of us (R. B. G.) thanks ...”. Instead, try “R. B. G. thanks...”. Put sponsor acknowledgments in the unnumbered footnote on the first page.

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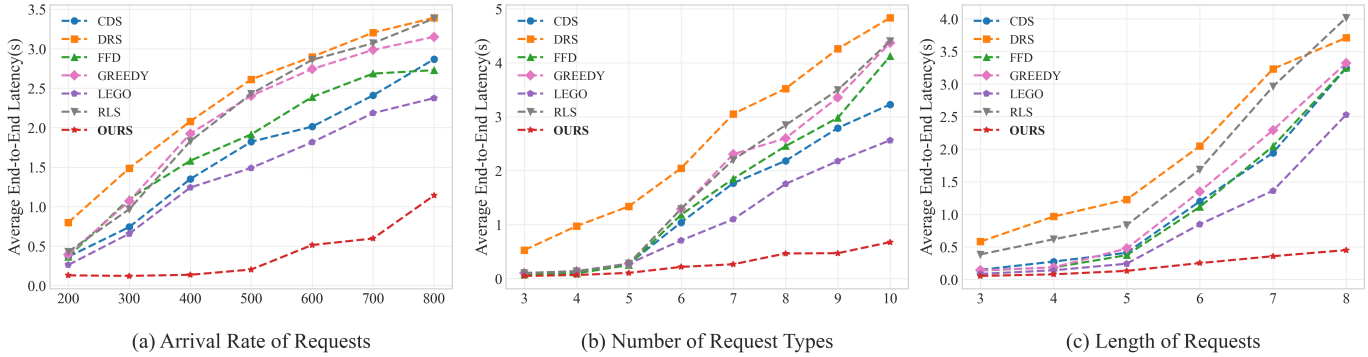

Fig. 2. Average end-to-end latency under three perturbation axes: (a) arrival rate, (b) number of request types, and (c) request chain length.

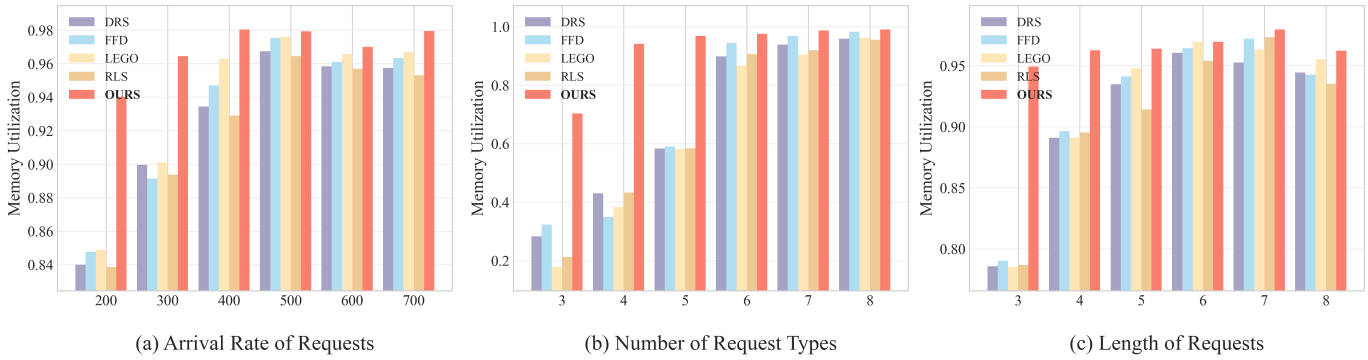

Fig. 3. Average memory utilization under three perturbation axes: (a) arrival rate, (b) number of request types, and (c) request chain length.

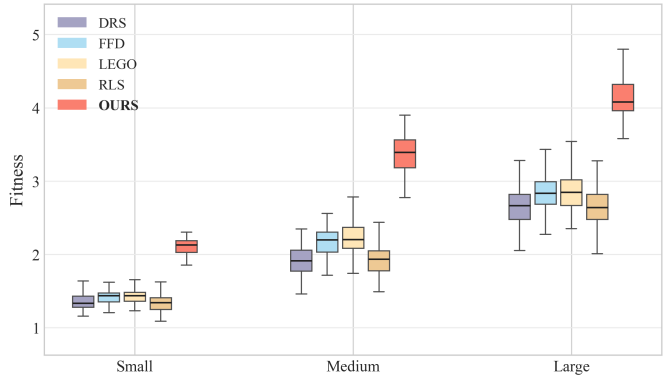

Fig. 4. Fitness distribution under three network scales (Small: $N = 3$, Medium: $N = 5$, Large: $N = 7$). Each box represents 100 independent trials.

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